



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

ASSIGNMENT No: 7

TITLE: Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement:

Implement a Java program to demonstrate inheritance and interfaces using suitable realworld examples.

Learning Objectives:

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory:

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Conclusion:

The Java application was successfully implemented using inheritance and interfaces. Inheritance was used to reuse common properties and methods, reducing code redundancy, while interfaces provided abstraction by defining a common contract for different classes. The Shape application and E-Commerce Product System demonstrated the practical implementation of object-oriented programming concepts such as code reusability, abstraction, and polymorphism, resulting in a well-structured and maintainable application.

Exercise

1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

```
import java.util.Scanner;
```

```
class Shape {  
    void area() {  
    }  
}
```

```
class Circle extends Shape {  
    double radius;
```

```
    Circle(double radius) {  
        this.radius = radius;  
    }
```

```
    void area() {  
        System.out.println("Area of Circle: " + (3.14 * radius * radius));  
    }  
}
```

```
class Rectangle extends Shape {  
    double length, width;
```

```
    Rectangle(double length, double width) {
```

```
    this.length = length;  
    this.width = width;  
}
```

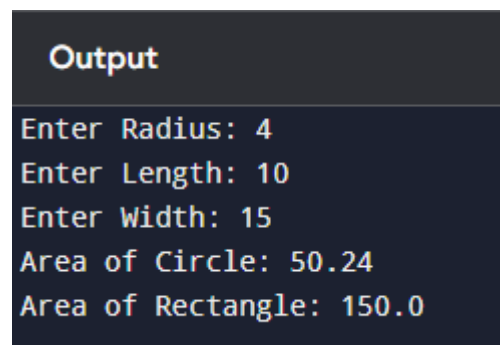
```
void area() {  
    System.out.println("Area of Rectangle: " + (length * width));  
}  
}
```

```
class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter Radius: ");  
        double r = sc.nextDouble();  
  
        System.out.print("Enter Length: ");  
        double l = sc.nextDouble();  
  
        System.out.print("Enter Width: ");  
        double w = sc.nextDouble();  
  
        Circle c = new Circle(r);  
        Rectangle rect = new Rectangle(l, w);  
  
        c.area();
```

```
        rect.area();

        sc.close();
    }
}
```

OUTPUT

A screenshot of a Java IDE's output window. The window has a dark background with a title bar that says "Output" in orange text. Below the title bar, the output text is displayed in a monospaced font with syntax highlighting: "Enter Radius: 4" (blue), "Enter Length: 10" (blue), "Enter Width: 15" (blue), "Area of Circle: 50.24" (blue), and "Area of Rectangle: 150.0" (blue).

```
Output
Enter Radius: 4
Enter Length: 10
Enter Width: 15
Area of Circle: 50.24
Area of Rectangle: 150.0
```

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```
interface Product {
    void display();
}
```

```
class ProductDetails {
    int id;
    String name;
    double price;

    ProductDetails(int id, String name, double price) {
```

```
        this.id = id;
        this.name = name;
        this.price = price;
    }
}
```

```
class Electronic extends ProductDetails implements Product {
```

```
    Electronic(int id, String name, double price) {
        super(id, name, price);
    }
```

```
    public void display() {
        System.out.println("Electronic Product");
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Price: " + price);
    }
}
```

```
class Clothing extends ProductDetails implements Product {
```

```
    Clothing(int id, String name, double price) {
        super(id, name, price);
    }
```

```
    public void display() {
        System.out.println("Clothing Product");
    }
```

```
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Price: " + price);
    }
}
```

```
class Grocery extends ProductDetails implements Product {
```

```
    Grocery(int id, String name, double price) {
        super(id, name, price);
    }
```

```
    public void display() {
        System.out.println("Grocery Product");
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Price: " + price);
    }
}
```

```
class Main {
```

```
    public static void main(String[] args) {
        Electronic e = new Electronic(101, "Laptop", 65000);
        Clothing c = new Clothing(102, "T-Shirt", 1200);
        Grocery g = new Grocery(103, "Rice", 850);

        e.display();
```

```
System.out.println();
```

```
c.display();
```

```
System.out.println();
```

```
g.display();
```

```
}
```

```
}
```

OUTPUT

Output

Electronic Product

ID: 101

Name: Laptop

Price: 65000.0

Clothing Product

ID: 102

Name: T-Shirt

Price: 1200.0

Grocery Product

ID: 103

Name: Rice

Price: 850.0